WORKED SOLUTION

STEP III 2012, Question 4

Pure • Summing Infinite Series

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/12-s3-q4

The key idea

Everything rests on two standard series: $\sum x^n/n! = \mathrm{e}^x$ and $\sum x^n/n = -\ln(1-x)$. In part (i) the trick is to write each polynomial in n so that it cancels against $n!$, which means using products like $n(n-1)$ and $n(n-1)(n-2)$ rather than powers of n. In part (ii), partial fractions turn the awkward rational factor into pieces of the form $1/(n+k)$, and each of those is a shifted logarithm series.

Solution

Part (i)

Two facts do all the work. First, $\sum_{n=0}^{\infty} 1/n! = \mathrm{e}$, so

$$\sum_{n=1}^{\infty} \frac{1}{n!} = \mathrm{e} - 1.$$

Second, for any whole number $k \geq 1$, the term $n(n-1)\cdots(n-k+1)/n!$ is zero when $n < k$ and equals $1/(n-k)!$ when $n \geq k$. So

$$\sum_{n=1}^{\infty} \frac{n(n-1)\cdots(n-k+1)}{n!} = \sum_{m=0}^{\infty} \frac{1}{m!} = \mathrm{e} \qquad (k \geq 1). \tag{$*$}$$

All the series here converge absolutely (each is a polynomial in n over $n!$), so we may split them up and recombine them freely.

First sum. Since $n+1 = n+1$, using $(*)$ with $k = 1$,

$$\sum_{n=1}^{\infty} \frac{n+1}{n!} = \mathrm{e} + (\mathrm{e} - 1) = 2\mathrm{e} - 1,$$

as required.

Second sum. We write $(n+1)^2 = n^2+2n+1 = n(n-1)+3n+1$. Then $(*)$ with $k=2$ and $k=1$ gives

$$\sum_{n=1}^{\infty} \frac{(n+1)^2}{n!} = \mathrm{e} + 3\mathrm{e} + (\mathrm{e}-1) = 5\mathrm{e}-1,$$

as required.

Third sum.

Ask yourself How do we turn $(2n-1)^3$ into falling products $n(n-1)(n-2)$, $n(n-1)$, n?

Expanding gives $(2n-1)^3 = 8n^3 - 12n^2 + 6n - 1$. Next we use

$$n^3 = n(n-1)(n-2) + 3n(n-1) + n, \qquad n^2 = n(n-1) + n.$$

You can check these by expanding the right-hand sides, or by putting in $n = 0, 1, 2, 3$. Substituting,

$$\begin{aligned}(2n-1)^3 &= 8[n(n-1)(n-2) + 3n(n-1) + n] - 12[n(n-1)+n] + 6n - 1 \\ &= 8\,n(n-1)(n-2) + 12\,n(n-1) + 2n - 1.\end{aligned}$$

(A quick test: at $n=1$ both sides are 1, and at $n=2$ both are 27.) Now $(*)$ with $k = 3, 2, 1$ gives

$$\sum_{n=1}^{\infty} \frac{(2n-1)^3}{n!} = 8\mathrm{e} + 12\mathrm{e} + 2\mathrm{e} - (\mathrm{e}-1) = 21\mathrm{e} + 1.$$

Part (ii)

Partial fractions. The numerator and denominator have the same degree, so we divide first:

$$\frac{n^2+1}{(n+1)(n+2)} = 1 - \frac{3n+1}{(n+1)(n+2)}.$$

Writing $\dfrac{3n+1}{(n+1)(n+2)} = \dfrac{A}{n+1} + \dfrac{B}{n+2}$, the cover-up rule gives $A = \dfrac{-2}{1} = -2$ (at $n=-1$) and $B = \dfrac{-5}{-1} = 5$ (at $n=-2$). Hence

$$\frac{n^2+1}{(n+1)(n+2)} = 1 + \frac{2}{n+1} - \frac{5}{n+2}.$$

The three pieces. We put $x = \frac{1}{2}$ and use $-\ln(1-x) = \sum_{m=1}^{\infty} x^m/m$, which holds for $|x| < 1$. Each of the three series below converges (the terms shrink at least as fast as

2^{-n}), so splitting the sum is legitimate.

$$\sum_{n=0}^{\infty} x^n = \frac{1}{1-x} = 2,$$
$$\sum_{n=0}^{\infty} \frac{x^n}{n+1} = \frac{1}{x}\sum_{m=1}^{\infty} \frac{x^m}{m} = \frac{-\ln(1-x)}{x} = 2\ln 2,$$
$$\sum_{n=0}^{\infty} \frac{x^n}{n+2} = \frac{1}{x^2}\sum_{m=2}^{\infty} \frac{x^m}{m} = \frac{-\ln(1-x)-x}{x^2} = 4(\ln 2 - \tfrac{1}{2}) = 4\ln 2 - 2.$$

In the second and third lines we put $m = n+1$ and $m = n+2$. The third line is the full logarithm series with its $m = 1$ term, x, taken away.

? **Ask yourself** Why does the third piece need the extra $-x$?

The shift $m = n+2$ makes the sum start at $m = 2$, so the series is $-\ln(1-x)$ with its first term missing. Forgetting this is the easiest way to go wrong here.

Combining.

$$\sum_{n=0}^{\infty} \frac{(n^2+1)2^{-n}}{(n+1)(n+2)} = 2 + 2(2\ln 2) - 5(4\ln 2 - 2) = 12 - 16\ln 2.$$

Checks

- Partial sums up to $n = 60$ give 4.436564, 12.591409 and 58.083918 for the three sums in (i). These match $2e-1$, $5e-1$ and $21e+1$ to machine precision.
- In (ii), $12 - 16\ln 2 \approx 0.909645$, which agrees with the partial sums. The first few terms are $\frac{1}{2}$, $\frac{1}{6}$, $\frac{5}{48}$, $\frac{1}{16}$, so the total is plausibly a little under 1.
- The answer to (ii) must be positive, and it is: $16\ln 2 \approx 11.09 < 12$.

Common slips

- Using $\sum_{n=1}^{\infty} 1/n! = e$, which forgets that the $n = 0$ term is missing.
- Writing $\sum n^2/n!$ as $\sum n/(n-1)!$ and then stopping there. You still have to split $n = (n-1)+1$.
- In (ii), dropping the $-x$ when shifting the index on $\sum x^n/(n+2)$.

Beyond STEP

The falling products $n(n-1)\cdots(n-k+1)$ are called falling factorials, and writing a polynomial in terms of them uses the Stirling numbers of the second kind. The sum in

$(*)$ says that the kth factorial moment of a Poisson distribution with mean 1 is 1. This is why, for example, $\sum n^k/n!$ is e times the kth Bell number. The rearrangements here are justified by absolute convergence, a key theme in first-year analysis: see Abbott, *Understanding Analysis*. For the probability side, see Blitzstein and Hwang, *Introduction to Probability*.

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